

Quiz04 - Query Language - 24A01

Total points **6/10** ?

The respondent's email (tphuc2438@apcs.fitus.edu.vn) was recorded on submission of this form.

0 of 0 points

StudentID *

24125076

Question 1

1 of 1 points

✓ Given the initial table R and the result table S. Which of the following expressions does NOT give the correct result? *1/1

R	A	B	C	D
	a1	b1	5	40
	a2	b2	4	38
	a3	b1	2	43
	a4	b1	1	55
	a5	b1	4	43

S	A	B	C	D
	a1	b1	5	40
	a5	b1	4	43

$\sigma_{(B=b1 \wedge C \geq 4)}(R)$

☐ Option 3

$\sigma_{(C=5 \vee C=4)}(\sigma_{D \geq 40}(R))$

☐ Option 1

$\sigma_{(B=b1 \wedge D \geq 40)}(R)$

☒ Option 2



$\sigma_{(D \geq 40)}(\sigma_{(C=5 \vee C=4)}(R))$

☐ Option 4

Question 2

0 of 1 points

✗ Given table Physician (ID, name, birthday) with 5 rows and Checkup (physicianID, patientID, date) with 7 rows.
How many rows will be a result when join condition is ID=physicianID? *0/1

☒ 5

✗

☐ 25☐ 12☐ 7

Question 3

0 of 1 points

✗ Given R(A, B, C). Which of the following formulas has a logical value regardless of substituting the actual value of the tuple into the tuple variable? *0/1

$$t \in R$$
☐ Option 1
$$t.B > 10$$
☐ Option 2
$$t.B < t.C$$
☐ Option 3☒ All of the above

✗

Question 4

1 of 1 points

✓ Given two relations R and S. Which of the following sentences is correct *1/1
when R naturally joins S?

- ☐ R and S do not need a compatible attribute type
- ☒ R and S must have at least one similar attribute ✓
- ☐ R and S must be identical
- ☐ R and S necessarily have the same number of attributes

Question 5

1 of 1 points

✓ Which of the following statements is NOT appropriate about join operations? *1/1

- ☐ The result of a natural join has fewer attributes than the result of an equijoin
- ☐ The results of natural and equi join are the same in terms of number of tuples
- ☐ Natural join is a special case of equi join
- ☒ The result of a natural join has more attributes than the result of an equijoin ✓

Question 6

1 of 1 points

✓ $\text{Stu}(\underline{sName}), \text{Reg}(\underline{sName}, \underline{cName}, \text{Grade})$. Choose the correct formula to find students who register DB course with grade>9. *1/1

- ☒ All formulas are incorrect ✓
- ☐ $\{t.sName \mid \forall t \in \text{Reg} \wedge t.cName = \text{'DB'} \wedge \exists q \in \text{Stu} (q.sName = t.cName \wedge t.Grade > 9)\}$
- ☐ $\{t.sName \mid t \in \text{Reg} \wedge t.cName = \text{'DB'} \wedge \forall q \in \text{Stu} (t.Grade > 9)\}$
- ☐ $\{t.sName \mid t \in \text{Reg} \wedge t.cName = \text{'DB'} \wedge \forall q \in \text{Stu} (q.sName = t.cName \wedge t.Grade > 9)\}$

Question 7

1 of 1 points

✓ Song(ID, Composer), Translation(ID, Name, Translator). Which of the following formulas is valid?

1/1

- ☐ $\{ t \mid t \in \text{Song} \wedge (\exists q \in \text{Translation} (t.\text{ID} = q.\text{ID} \wedge t.\text{Composer} = \text{'Bryan'})) \}$
- ☐ $\{ s \mid s \in \text{Song} \wedge \neg(\exists q \in \text{Translation} (s.\text{ID} = q.\text{ID})) \}$
- ☐ $\{ v \mid v \in \text{Song} \wedge \neg(v.\text{Composer} = \text{'Bryan'}) \}$

☒ All formulas are valid



Question 8

1 of 1 points

✓ Given a database & query as figure below. Which of the following English *1/1 sentences describes the result of the query?

College(cName, state, enrollment)
Student(sID, sName, GPA, sizeHS)
Apply(sID, cName, major, decision)

```
SELECT s.sName, c.cName
FROM Student s, College c, Apply a
WHERE s.sID=a.sID AND c.cName=a.cName
AND s.sizeHS>c.enrollment
```

- ☒ Find all student-college name pairs, where the college to which students applied is smaller than their high school
- ☐ Find all student-college name pairs, where colleges are smaller than the student's high school
- ☐ Find all student-college name pairs, where the college is smaller than some high school
- ☐ None of them is correct



Question 9

0 of 1 points

✗ Given a database as figure below. Which of the following expressions finds the IDs of all students such that some college bears the student's name? *0/1

College(cName, state, enrollment)

Student(sID, sName, GPA, sizeHS)

Apply(sID, cName, major, decision)

$\pi_{sID}(\pi_{cName}College \bowtie \pi_{cName}(\sigma_{sName=cName}Student))$

$\pi_{sID}(\sigma_{cName=sName}(\pi_{sID}Student \times College \times Student))$

☐ Option 3

☐ Option 4

$\pi_{sID}(College \bowtie Student)$

$\pi_{sID}(\sigma_{cName=sName}(College \times \pi_{cName}Apply))$

☒ Option 1

✗

☐ Option 2

Question 10

0 of 1 points

✗ Given a database as figure below. Suppose relation Student has 20 tuples. What is the minimum and maximum number of tuples in the result of the expression? *0/1

College(cName, state, enrollment)

Student(sID, sName, GPA, sizeHS)

Apply(sID, cName, major, decision)

$\rho_{s1(i1,n1,g,h)} Student \bowtie \rho_{s2(i2,n2,g,h)} Student$

- ☐ minimum = 0, maximum = 400
- ☐ minimum = 20, maximum = 400
- ☐ minimum = 40, maximum = 40
- ☒ minimum = 20, maximum = 20



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